

Problem

The flash unit of a camera has a 2-mF capacitor charged to 40 V.

(a) How much charge is on the capacitor?

(b) What is the energy stored in the capacitor?

(c) If the flash fires in 0.8 ms, what is the average current through the flashtube?

(d) How much power is delivered to the flashtube? After a picture has been taken, the capacitor needs to be recharged by a power unit that supplies a maximum of 5 mA. How much time does it take to charge the capacitor?

Step-by-step solution

Step 1 of 6

(a)

Write the expression for charge on the capacitor.

$$Q = CV_s$$

Substitute 2 mF for C and 40 V for V_s in the equation.

$$\begin{aligned} Q &= (2 \text{ mF})(40 \text{ V}) \\ &= 80 \text{ mC} \end{aligned}$$

Therefore, the charge, Q on the capacitor is 80 mC.

Step 2 of 6

(b)

Write the expression for energy stored in a capacitor.

$$W = \frac{1}{2} CV_s^2$$

Substitute 2 mF for C and 40 V for V_s in the equation.

$$\begin{aligned} W &= \frac{1}{2} (2 \times 10^{-3}) (40)^2 \\ &= 1.6 \text{ J} \end{aligned}$$

Therefore, the energy, W stored in the capacitor is 1.6 J.

Step 3 of 6

(c)

The flash fires for every time constant (τ).

$$\tau = RC$$

Substitute 2 mF for C and 0.8 ms for τ in the equation.

$$\begin{aligned} R &= \frac{0.8(10^{-3})}{2(10^{-3})} \\ &= 0.4\Omega \end{aligned}$$

Determine the current through the flash tube.

$$\begin{aligned} I &= \frac{V_s}{R} \\ &= \frac{40}{0.4} \\ &= 100\text{ A} \end{aligned}$$

Therefore, the average current through the flash tube is $\boxed{100\text{ A}}$.

Step 4 of 6

(d)

The power delivered to the flash tube is,

$$p = \frac{W}{\tau}$$

Substitute 1.6 J for W and 0.8 ms for τ in the equation.

$$\begin{aligned} p &= \frac{1.6}{0.8(10^{-3})} \\ &= 2\text{ kW} \end{aligned}$$

Therefore, the power, p delivered to the flash tube is $\boxed{2\text{ kW}}$.

Step 5 of 6

The peak charging current is,

$$\begin{aligned} I &= \frac{V_s}{R} \\ R &= \frac{V_s}{I} \end{aligned}$$

Substitute 40 V for V_s and 5 mA for I in the equation.

$$\begin{aligned} R &= \frac{40}{5(10^{-3})} \\ &= 8\text{ k}\Omega \end{aligned}$$

Step 6 of 6

The time taken to charge the capacitor is,

$$\begin{aligned}\tau &= RC \\ &= (8 \times 10^3)(2 \times 10^{-3}) \\ &= 16 \text{ s}\end{aligned}$$

Therefore, the time, τ taken to charge the capacitor is 16s.